

Lab Program 01: Implement Tic Tac Toe

```
board = [' ']*9
```

```
def display():
```

```
    print()
```

```
    print(board[0], "|", board[1], "|", board[2])
```

```
    print("--+---+--")
```

```
    print(board[3], "|", board[4], "|", board[5])
```

```
    print("--+---+--")
```

```
    print(board[6], "|", board[7], "|", board[8])
```

```
def win(p):
```

```
    return ((board[0]==board[1]==board[2]==p) or
```

```
            (board[3]==board[4]==board[5]==p) or
```

```
            (board[6]==board[7]==board[8]==p) or
```

```
            (board[0]==board[3]==board[6]==p) or
```

```
            (board[1]==board[4]==board[7]==p) or
```

```
            (board[2]==board[5]==board[8]==p) or
```

```
            (board[0]==board[4]==board[8]==p) or
```

```
            (board[2]==board[4]==board[6]==p))
```

```
for turn in range(9):
```

```
    display()
```

```
if turn % 2 == 0:

    move = int(input("Enter position (1-9): "))

    if board[move-1] in ['X', 'O']:

        print("Position already taken!")

        continue

    board[move-1] = 'X'

    if win('X'):

        display()

        print("You win!")

        break

else:

    print("Computer's turn...")

    for i in range(9):

        if board[i] not in ['X', 'O']:

            board[i] = 'O'

            break

    if win('O'):

        display()

        print("Computer wins!")

        break

else:

    display()

    print("It's a draw!")
```

Output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/admin/Downloads/IBF24CS189_AI/TIC_TAC_TOE_183.py =====

  | |
  | |
  | |
Enter position (1-9): 5

  | |
  | X |
  | |
Computer's turn...

O | |
  | X |
  | |
Enter position (1-9): 7

O | |
  | X |
  | |
X | |
Computer's turn...

O | O |
  | X |
  | |
X | |
Enter position (1-9): 5
Position already taken!

O | O |
  | X |
  | |
X | |
Computer's turn...

O | O | O
  | X |
  | |
X | |
Computer wins!

>>>
>>> Murra Sai Keerthana - IBF24CS189
```

Activate Windows
Go to Settings to activate Windows.

Ln 57 Col 32